Solution

(i) Let $I = \sqrt{4-x^2}\,dx$

$$= \int \sqrt{2^2-x^2}\,dx$$

$$= \frac{x}{2}\sqrt{2^2-x^2} + \frac{2^2}{2}\sin^{-1}\left(\frac{x}{2}\right) + c$$

Therefore, $I = \frac{x}{2}\sqrt{4-x^2} + 2\sin^{-1}\left(\frac{x}{2}\right) + c$

(ii) Let $I = \int \sqrt{25x^2-9}\,dx$

$$= \int \sqrt{(5x)^2-3^2}\,dx$$

$$= \frac{1}{5}\left[\frac{5x}{2}\sqrt{(5x)^2-3^2} - \frac{3^2}{2}\log\left|5x+\sqrt{(5x)^2-3^2}\right|\right] + c$$

Therefore, $I = \frac{1}{5}\left[\frac{5x}{2}\sqrt{25x^2-9} - \frac{9}{2}\log\left|5x+\sqrt{25x^2-9}\right|\right] + c$

(iii) Let $I = \int \sqrt{x^2+x+1}\,dx$

$$= \int \sqrt{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\,dx$$

$$= \frac{x+\frac{1}{2}}{2}\sqrt{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} + \frac{\left(\frac{\sqrt{3}}{2}\right)^2}{2}\log\left|\left[x+\frac{1}{2}+\sqrt{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right]\right| + c$$

Therefore, $I = \frac{2x+1}{4}\sqrt{x^2+x+1} + \frac{3}{8}\log\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right| + c$

(iv) Let $I = \int \sqrt{(x-3)(5-x)}\,dx$

$$= \int \sqrt{8x-x^2-15}\,dx$$

$$= \int \sqrt{1^2-(x-4)^2}\,dx$$

$$= \frac{x-4}{2}\sqrt{1^2-(x-4)^2} + \frac{1}{2}\sin^{-1}\left(\frac{x-4}{1}\right) + c$$

Therefore, $I = \frac{x-4}{2}\sqrt{8x-x^2-15} + \frac{1}{2}\sin^{-1}(x-4) + c$

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